ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Section A: objective questions (30) · Section B: industrial sensors and their connection, PLCs and ladder logic with timers and counters, power transmission (belts, chains, torque and power), bearings and couplings, and the design of a sorting system (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. Use π = 3,142 where needed.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A proximity sensor that detects only metal objects is a/an …

A capacitive sensor. B ultrasonic sensor.
C LDR. D inductive sensor.

1.1.2 A sensor that detects plastic, wood and liquids without touching them is a/an …

A inductive sensor. B capacitive sensor.
C reed switch. D thermocouple.

1.1.3 A photo-electric sensor where the emitter and receiver are in separate housings facing each other is a …

A diffuse sensor. B retro-reflective sensor.
C through-beam sensor. D proximity switch.

1.1.4 A PNP sensor output, when switched on, connects the load to …

A the positive supply. B 0 V.
C earth only. D the PLC's CPU.

1.1.5 In ladder logic, a contact that is closed when its input is OFF is drawn as a …

A normally open contact. B normally closed contact.
C coil. D timer.

1.1.6 A timer that turns its output on a set time after its input turns on is an …

A off-delay timer. B up-counter. C latch. D on-delay timer.

1.1.7 Belt drives are used instead of gears when …

A the shafts are far apart and quiet running is needed.
B an exact speed ratio is essential.
C very high torque must be transmitted without slip.
D shafts must cross at 90° only.

1.1.8 Power transmitted by a rotating shaft is calculated with …

A $P = F \times A$
B $P = V \div R$
C $P = 2\pi NT \div 60$
D $P = m \times g$

1.1.9 A bearing that uses balls between an inner and outer ring is a …

A plain bearing.
B bush.
C rolling-element bearing.
D thrust washer.

1.1.10 A coupling that allows for small misalignment between two shafts is a …

A flexible coupling.
B rigid coupling.
C keyway.
D circlip.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 An inductive sensor can detect an aluminium can.

1.2.2 The sensing distance of an inductive sensor is larger for steel than for brass.

1.2.3 An NPN sensor is also called a sinking output.

1.2.4 A PLC scans its program continuously: read inputs, execute program, update outputs.

1.2.5 A counter's output turns on when the count reaches its preset value.

1.2.6 Chain drives can slip like flat belts.

1.2.7 Toothed (timing) belts give a fixed speed ratio without slip.

1.2.8 Bearings must be lubricated to reduce friction and wear.

1.2.9 Shaft misalignment can cause vibration and early bearing failure.

1.2.10 An emergency stop button should be wired as a normally open contact.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Distance at which a proximity sensor switches	A	key
1.3.2	Sensor that reflects light off a special reflector	B	reed switch
1.3.3	Magnetically operated switch in a glass tube	C	torque
1.3.4	PLC instruction that keeps an output on after the start button is released	D	thrust bearing
1.3.5	Time for one complete PLC program cycle	E	latch / seal-in
1.3.6	Value at which a timer or counter switches	F	velocity ratio

1.3.7	Diameter of driven pulley ÷ diameter of driver pulley	G	sensing range
1.3.8	Turning force on a shaft	H	scan time
1.3.9	Bearing that supports a load along the shaft's axis	I	preset
1.3.10	Small metal piece that locks a pulley to a shaft	J	retro-reflective

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: INDUSTRIAL SENSORS

2.1 Recommend a sensor for EACH task and give a reason: (a) detecting steel bolts on a conveyor; (b) checking the level of rice in a hopper; (c) detecting cardboard boxes 3 m away; (d) confirming that a pneumatic cylinder is fully extended. (8)

2.2 Explain the difference between an NPN (sinking) and a PNP (sourcing) sensor output, and draw how a PNP sensor is connected to a PLC input. (6)

2.3 Give THREE factors that affect the sensing distance of a proximity sensor. (3)

[17]

QUESTION 3: PLCs AND LADDER LOGIC

3.1 Explain the PLC scan cycle. (3)

3.2 A conveyor must run when START (I0.0) is pressed and stop when STOP (I0.1) is pressed. A warning horn (Q0.1) must sound for 5 seconds before the conveyor motor (Q0.0) starts. Draw the ladder diagram using an on-delay timer T1. (8)

3.3 A packing machine must count 12 bottles (sensor I0.2) and then switch on a box-change light (Q0.2). The light and counter reset with a button (I0.3). Draw the ladder rungs. (5)

[16]

QUESTION 4: POWER TRANSMISSION, BEARINGS AND COUPLINGS

4.1 A motor pulley of 120 mm diameter turns at 1 440 r/min and drives a pulley of 360 mm diameter with a V-belt.

4.1.1 Calculate the speed of the driven pulley. (3)

4.1.2 The driven shaft delivers a torque of 30 N·m. Calculate the power at the driven shaft. (3)

4.2 Compare belt drives and chain drives under the headings slip, noise, lubrication and distance between shafts. (4)

4.3 Name THREE causes of bearing failure and give ONE sign that a bearing is failing. (4)

4.4 Explain why two shafts joined by a coupling must be aligned, and name ONE method of checking alignment. (3)

[17]

QUESTION 5: DESIGN OF A SORTING SYSTEM

5.1 A machine must sort metal and plastic parts on a conveyor. Metal parts must be pushed off the belt into a bin by a pneumatic cylinder; plastic parts carry on to the end.

5.1.1 Name the sensors you would use and the purpose of each. (4)

5.1.2 List the inputs and outputs of the PLC for this machine. (5)

5.1.3 Describe, step by step, how the machine must operate. (6)

5.1.4 Name TWO safety features the machine must have. (2)

5.1.5 Name ONE preventive maintenance task for the conveyor and ONE for the pneumatic system. (3)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER